



OASIS HRMS

ENTERPRISE SYSTEM BLUEPRINT

COMPLETE PLATFORM ARCHITECTURE & USER MANUAL

Human Resource Management System (v2.0)

A comprehensive guide detailing all functional modules, screens, user roles, database architecture, workflows, smart geofencing, ATS recruitment pipelines, and AI workforce capabilities.

3

USER ROLES
(ADMIN, HR, EMP)

16+

CORE HR & OPS
MODULES

100%

GEOFENCE +
BIOMETRICS

AI

OVI GEMINI VOICE
ASSISTANT

Prepared for: Subedge Technology &
Enterprise Clients

Stack: React Native / Expo 54 + Supabase
PostgreSQL

Date: 2026 Edition •
Confidential

1. Executive Summary & Tech Architecture

Oasis HRMS is an enterprise-grade, multi-tenant Human Capital Management & Workforce Automation platform engineered with a single unified codebase supporting **Web, Android, and iOS**. Built on **React Native + Expo SDK** and powered by **Supabase PostgreSQL** with Row-Level Security (RLS) and real-time synchronization.

Frontend Engine

Expo Router (file-based), React 19, TypeScript, React Native Web, Victory Native charts, Skia graphics.

Cloud & Database

Supabase PostgreSQL 15, Row-Level Security (RLS), Realtime WebSocket feeds, Edge Storage buckets.

AI & Smart Services

Google Gemini AI ("Ovi Voice Assistant"), Haversine GPS Geofencing, Facial biometrics, Resend Email API.

Role-Based Navigation & Access Matrix

The system dynamically adapts its interface, sidebar navigation, bottom tab bar, and route guards based on the authenticated user's role:

MODULE / FEATURE	ROUTE LOCATION	ADMIN	HR MANAGER	EMPLOYEE	PRIMARY PURPOSE
Admin Executive Console	<code>/(admin)/dashboard</code>	FULL	NO	NO	Organization KPIs, workforce analytics, payroll budget overview
HR Ops Console	<code>/(hr)/dashboard</code>	FULL	FULL	NO	Realtime attendance radar, pending approvals, quick actions
Employee Self-Service (ESS)	<code>/(employee)/dashboard</code>	VIEW	VIEW	FULL	1-tap GPS clock-in, leave quota cards, payslips, shift countdown
Users & Employee Master	<code>/(admin)/users</code>	FULL	FULL	NO	Staff directory, KYC, salary configuration, role assignment
Smart GPS Attendance	<code>/(hr)/attendance</code> & <code>/(employee)/attendance</code>	FULL	FULL	SELF	Geofence radius validation, biometric face snap, regularizations
Leave Management	<code>/(hr)/leave</code> & <code>/(employee)/leave</code>	FULL	FULL	APPLY	Leave quota balances, 1-click approvals,

MODULE / FEATURE	ROUTE LOCATION	ADMIN	HR MANAGER	EMPLOYEE	PRIMARY PURPOSE
& Workflow					holiday calendars
Enterprise ATS Recruitment	<code>/hr/recruitment</code> & <code>/careers</code>	FULL	FULL	NO	Job requisitions, pipeline Kanban, interview scoring, offer letters
Payroll & Payslip Engine	<code>/hr/payroll</code> & <code>/employee/payslips</code>	FULL	FULL	SELF	Salary structure, LOP calculation, PF/TDS, digital payslips
Performance & OKRs / 360	<code>/hr/performance</code> & <code>/employee/performance</code>	FULL	FULL	SELF	Goal milestones, self/manager appraisal reviews, peer Kudos
LMS & Learning Center	<code>/hr/learning</code> & <code>/employee/learning</code>	FULL	FULL	ENROLL	Course curricula, video lessons, quiz pass scores, certificates
IT Asset Management	<code>/hr/assets</code>	FULL	FULL	NO	Laptops, barcodes, warranty tracking, employee assignments
HR & IT Helpdesk	<code>/hr/helpdesk</code> & <code>/employee/helpdesk</code>	FULL	FULL	SUBMIT	Ticketing lifecycle, priority levels, threaded resolutions
Expense Claims & Receipts	<code>/hr/expenses</code> & <code>/employee/expenses</code>	FULL	FULL	SUBMIT	Receipt uploads, currency selection, manager reimbursement review
Shifts & Roster Management	<code>/hr/shifts</code>	FULL	FULL	VIEW	Morning/Evening/Night shifts, grace periods, roster assignment
AI Assistant ("Ovi")	<code>/employee/call-ovi</code>	FULL	FULL	FULL	Gemini AI voice interaction, policy answers, leave inquiries
Audit Trail & Compliance	<code>/admin/audit-logs</code>	FULL	NO	NO	Immutable audit log, user IP tracking, entity diff metadata

2. In-Depth Functional Modules (Where & How)

2.1 Smart GPS Geofence & Biometric Attendance

 **Location:** `/employee/attendance` (Employee) & `/hr/attendance` (HR/Admin)

Oasis prevents buddy-punching using a multi-factor attendance verification engine combining device GPS coordinates with server-side Haversine geodesic distance validation against workplace geofence perimeters, coupled with front-camera facial snapshot verification.

Employee Capabilities

- **1-Tap Smart Punch:** Instant Clock In and Clock Out with live GPS distance meter (e.g. "12m from Office 1 - Inside Geofence").
- **Face Recognition Snap:** Seamless front camera capture storing biometric verification logs.
- **Break Tracking:** Record tea, lunch, and custom breaks with accurate pause timers.
- **Attendance History:** Monthly calendar view indicating Present, Late, Half-day, Overtime, and Leave.
- **Regularization Requests:** Raise attendance adjustment requests with explanatory notes for forgotten punches.

HR & Admin Management

- **Organization Live Radar:** Monitor daily presence %, who is on-time, late, on leave, or absent in real-time.
- **Verification Logs:** Review GPS latitude/longitude coordinates and camera snapshot proof for every punch.
- **Workplace Office Geofencing:** Configure offices at `/hr/locations` with custom GPS coordinates and radius (e.g., 50m to 500m).
- **Shift Linkage:** Automatic shift grace period calculations (e.g., 15 mins allowance) before marking status as "Late".

2.2 Leave Management & Quota Automation

 **Location:** `/employee/leave` & `/hr/leave`

Automated entitlement management with dynamic quota tracking, multi-tier approvals, and automatic balance reconciliation.

Leave Categories Supported

- **Annual / Privilege Leave (AL):** 18 days default paid vacation.
- **Sick / Medical Leave (SL):** 12 days medical coverage.
- **Casual Leave (CL):** 7 days short personal leave.
- **Maternity / Paternity:** 90 days maternity / 7 days paternity.
- **Compensatory Off (Comp-Off):** 5 days for overtime/weekend shifts.

Workflows & Logic

- **Dynamic Balances:** Real-time deduction upon approval; instant refund upon cancellation of pending requests.
- **Half-Day Mode:** Allows First-Half or Second-Half 0.5-day leave applications.
- **HR Approval Queue:** Approve or Reject with mandatory manager remarks and automated email alert.
- **Holiday Calendar:** Integrates statutory public and regional holidays directly into the leave planner.

2.3 Payroll, Tax Deductions & Payslips

 **Location:** `/hr/payroll` & `/employee/payslips`

End-to-end compensation engine calculating gross pay, loss of pay (LOP) deductions, statutory deductions (PF, ESI, TDS), and digital payslip delivery.

PAYROLL COMPONENT	CALCULATION TYPE	DESCRIPTION
Basic Salary	Fixed Base (e.g. 50%)	Core taxable monthly salary defined per employee in staff master
House Rent Allowance (HRA)	Allowance (e.g. 40–50%)	Tax-exempt allowance under statutory guidelines
Special / Travel Allowances	Variable / Fixed	Medical, conveyance, night-shift allowances (calculated from shift logs)
Loss of Pay (LOP)	Automated Deduction	<code>(Basic Salary / Total Month Days) * Unapproved Absent Days</code>
Provident Fund (PF) & TDS	Statutory Deductions	12% PF contribution + withholding income tax calculations
Net Salary Disbursed	Gross Pay - Total Deductions	Final take-home pay generated with a unique digital Payslip ID

3. Talent, Learning & Operational Modules

3.1 Enterprise ATS & Recruitment Pipeline

 **Location:** [/\(hr\)/recruitment](/hr/recruitment) & Public Careers Portal </careers>

Full-cycle recruitment suite from job requisition creation to multi-channel publishing, Kanban applicant tracking, interview scorecards, and 1-click employee onboarding.

1 Job Requisition & Publishing

Create detailed job openings with salary ranges, experience, skills, perks, and target join date. Publish to Oasis Public Careers portal (</careers>), LinkedIn, Indeed, and Naukri.

2 Candidate Sourcing & Screening

External applicants apply via the public careers page. Resumes, portfolio links, and KYC are ingested into the ATS Kanban board across custom stages.

3 Interview Rounds & Structured Scoring

Schedule multi-round interviews (Technical, System Design, HR). Interviewers submit structured ratings (1–5 stars) and qualitative feedback.

4 Digital Offer Letter & Auto-Hire

Issue formal offer letters with compensation breakdown. Once accepted, HR clicks *"Convert to Employee"* to automatically generate their user account and staff profile.

3.2 Performance Management (OKRs & 360 Appraisals)

Goals & OKRs Tracking

[/\(hr\)/performance](/hr/performance) & [/\(employee\)/performance](/employee/performance)

Define Company, Departmental, and Individual Objectives & Key Results.

- Progress percentage sliders with measurable milestones.
- Target date tracking and priority flags (High, Medium, Low).
- Direct linkage to quarterly and annual appraisal cycles.

360 Review Cycles & Kudos

[Appraisal Reviews & Social Feed](#)

Holistic employee evaluations with peer recognition.

- **Self-Review & Rating:** Employees evaluate their own achievements.
- **Manager Review & Rating:** Supervisors provide scores & promotion recommendations.
- **Peer Kudos Badges:** Send social badges (e.g. *"Innovator"*, *"Team Player"*).

3.3 Learning Management System (LMS) & Training

Course Catalog & Curricula

`/(hr)/learning & /(employee)/learning`

- Organize training by department: Engineering, Leadership, Compliance, Security.
- Support for video lessons, rich markdown modules, and downloadable materials.
- Designate courses as **Mandatory** for mandatory annual compliance.

Progress & Digital Certificates

Automated Issuance

- Lesson completion tracking with live progress bar.
- Pass criteria evaluation (e.g., 80% passing grade).
- Automatic generation and download of branded completion certificates.

3.4 Hardware Asset Management & Helpdesk Ticketing

IT Asset Tracking (`/(hr)/assets`)

- Manage IT hardware (MacBooks, monitors, test devices).
- Barcode / Asset Tagging (e.g. `AST-MBP-2026-001`).
- Track serial numbers, purchase price, current book value, warranty expiry, and assigned staff member.

Internal Helpdesk (`/helpdesk`)

- Employees raise tickets for HR, IT, Payroll, or Facilities.
- Automatic ticket number generation (`TCK-8921`).
- Priority handling (Urgent, High, Medium, Low) with resolution notes and status tracking.

4. AI Assistant ("Ovi"), Security & Operations

4.1 Ovi AI – Google Gemini Powered Workforce Assistant

 **Location:** `/(employee)/call-ovi`

Oasis features an integrated conversational and voice AI assistant powered by **Google Gemini**. Ovi acts as an on-demand virtual HR specialist for all employees and managers.

Policy & FAQ Queries

Instantly answers company policies, leave entitlements, health insurance benefits, and reimbursement limits.

Leave & Attendance Inquiries

Provides real-time answers on remaining casual/sick leave days, past punch logs, and holiday lists.

Voice Call Experience

Interactive voice UI allowing hands-free employee self-service inquiries with audio waveform visualizations.

4.2 Enterprise Security, Governance & Audit Trails

Row-Level Security (RLS)

Every single table in PostgreSQL is protected by strict RLS policies ensuring complete multi-tenant and departmental isolation.

Comprehensive Audit Logs

`/(admin)/audit-logs`

Tracks all privileged actions (role changes, salary updates, geofence modifications) with actor ID, IP address, and JSON diffs.

4.3 Database Schema Quick Reference (Supabase PostgreSQL)

TABLE NAME	PRIMARY KEY	KEY RELATIONS / FIELDS	PURPOSE
<code>organizations</code>	<code>id</code> (UUID)	<code>name</code> , <code>slug</code> , <code>logo_url</code> , <code>package_type</code> , <code>settings</code>	Multi-tenant organizational accounts & configs
<code>profiles</code>	<code>id</code> (UUID)	<code>organization_id</code> , <code>full_name</code> , <code>email</code> , <code>role</code> , <code>avatar_url</code>	User authentication & role metadata
<code>employees</code>	<code>id</code> (UUID)	<code>profile_id</code> , <code>department_id</code> , <code>workplace_id</code> , <code>basic_salary</code>	Staff operational records & compensation info
<code>workplaces</code>	<code>id</code> (UUID)	<code>latitude</code> , <code>longitude</code> , <code>radius_meters</code> , <code>address</code>	Office GPS locations for geofenced clocking
<code>attendance</code>	<code>id</code> (TEXT)	<code>employee_id</code> , <code>date</code> , <code>clock_in</code> , <code>clock_out</code> , <code>face_verified</code>	Daily punch records, coordinates, biometric proof

TABLE NAME	PRIMARY KEY	KEY RELATIONS / FIELDS	PURPOSE
leave_requests	id (UUID)	employee_id, leave_type_id, start_date, status	Leave applications and manager approval states
payroll	id (UUID)	payroll_period_id, basic_salary, gross_salary, net_salary	Monthly salary calculations and payslips
assets	id (TEXT)	asset_tag, serial_number, assigned_to, value	Company hardware & IT equipment inventory

4.4 Default Test Accounts for Verification

ROLE	EMAIL	DEFAULT PASSWORD	ACCESSIBLE MODULES
Super Admin	ayushbindhani001@gmail.com / admin@oasis.local	Nanda@5152 / Admin@12345	Executive Dashboard, Audit Logs, Settings, Users, All HR Modules
HR Manager	hr@oasis.local	HR@12345	HR Dashboard, Recruitment/ATS, Payroll, Attendance, Leaves, Assets
Employee	employee@oasis.local	Employee@12345	Self-Service, 1-Tap Clock-in, Leave Apply, Payslips, LMS, Ovi AI

✔ Oasis HRMS Summary

This manual covers the complete functionality of Oasis HRMS. All features are accessible across Desktop Web browsers and Android / iOS devices with full cross-platform parity.